

Introduction to cancer claims

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What we'll cover

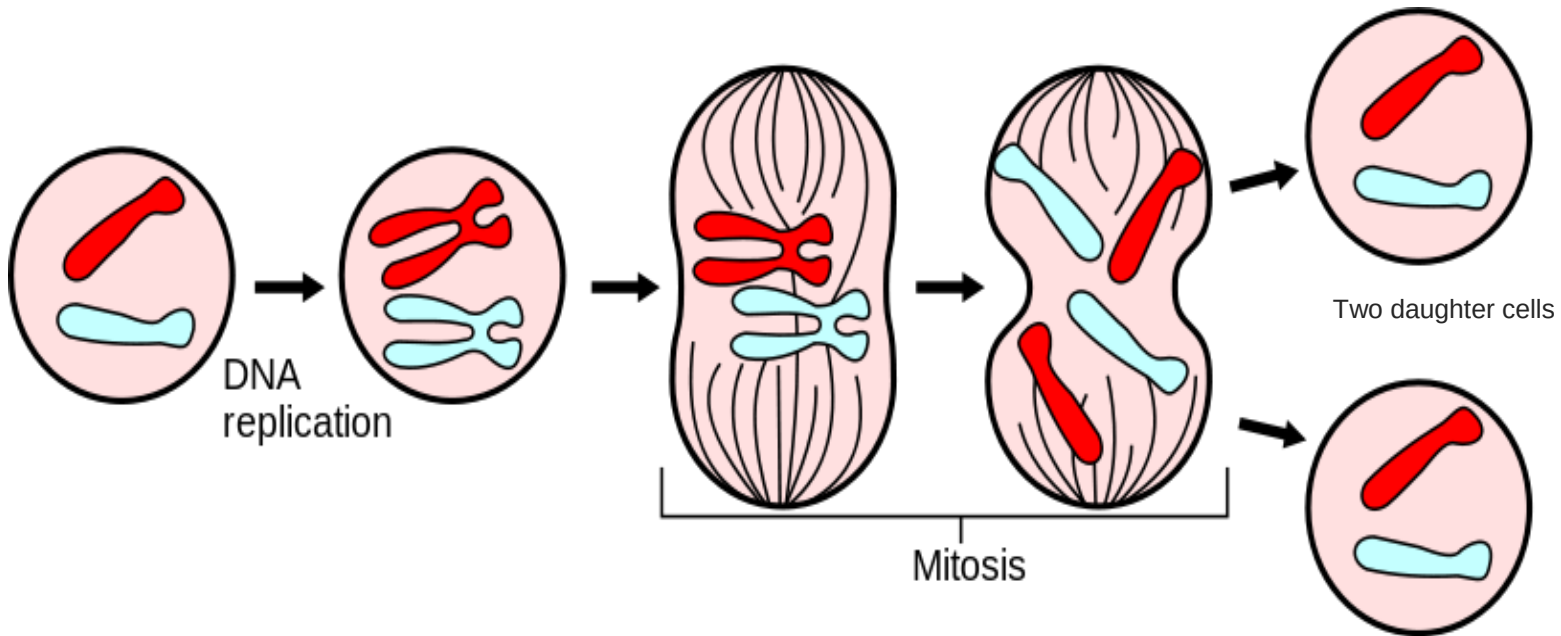
- Facts and figures
- What is cancer
- Risk factors and symptoms
- CI definition changes over time
- Assessing cancer claims
- Detailed look at breast, bowel and skin cancers
- Questions & Case studies (if time)

Facts and figures

- How many people in the UK are currently living with cancer?
- 2,500,000
- How many people are being diagnosed with cancer each day this year?
- 1,000
- What's the most common cancer in females?
- Breast cancer (31%)
- ..and males?
- Prostate cancer (26%)
- And the second most prevalent for both sexes?
- Bowel and lung cancers (13% each)

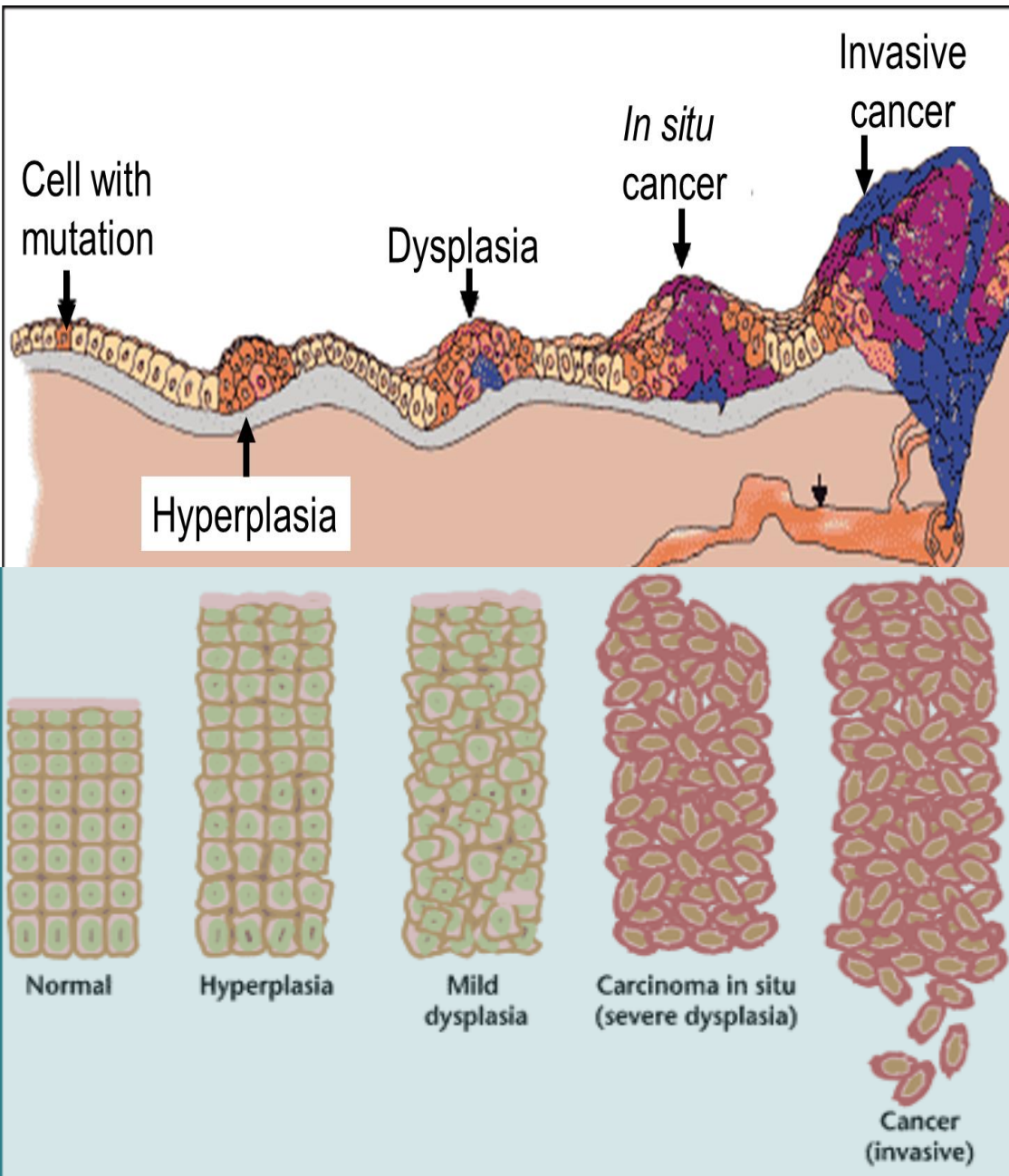
What is cancer?

Our body is made up of approx. 37 trillion individual cells



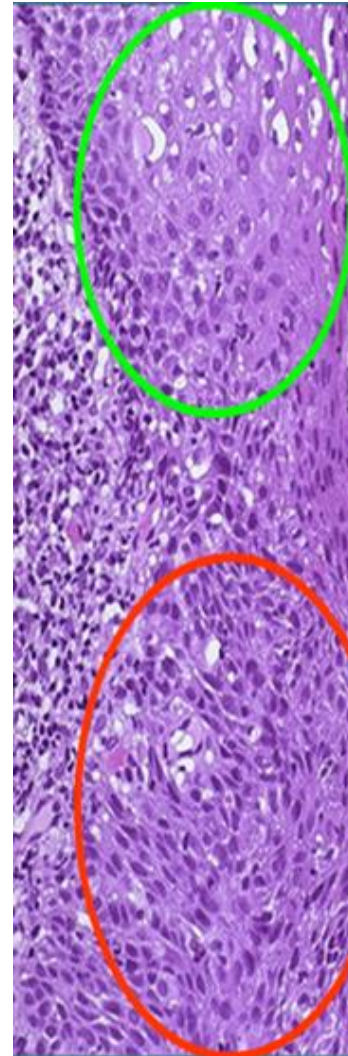
- These 2 daughter cells replace 1 original cell and 1 dying cell through a process called mitosis.

What is cancer?



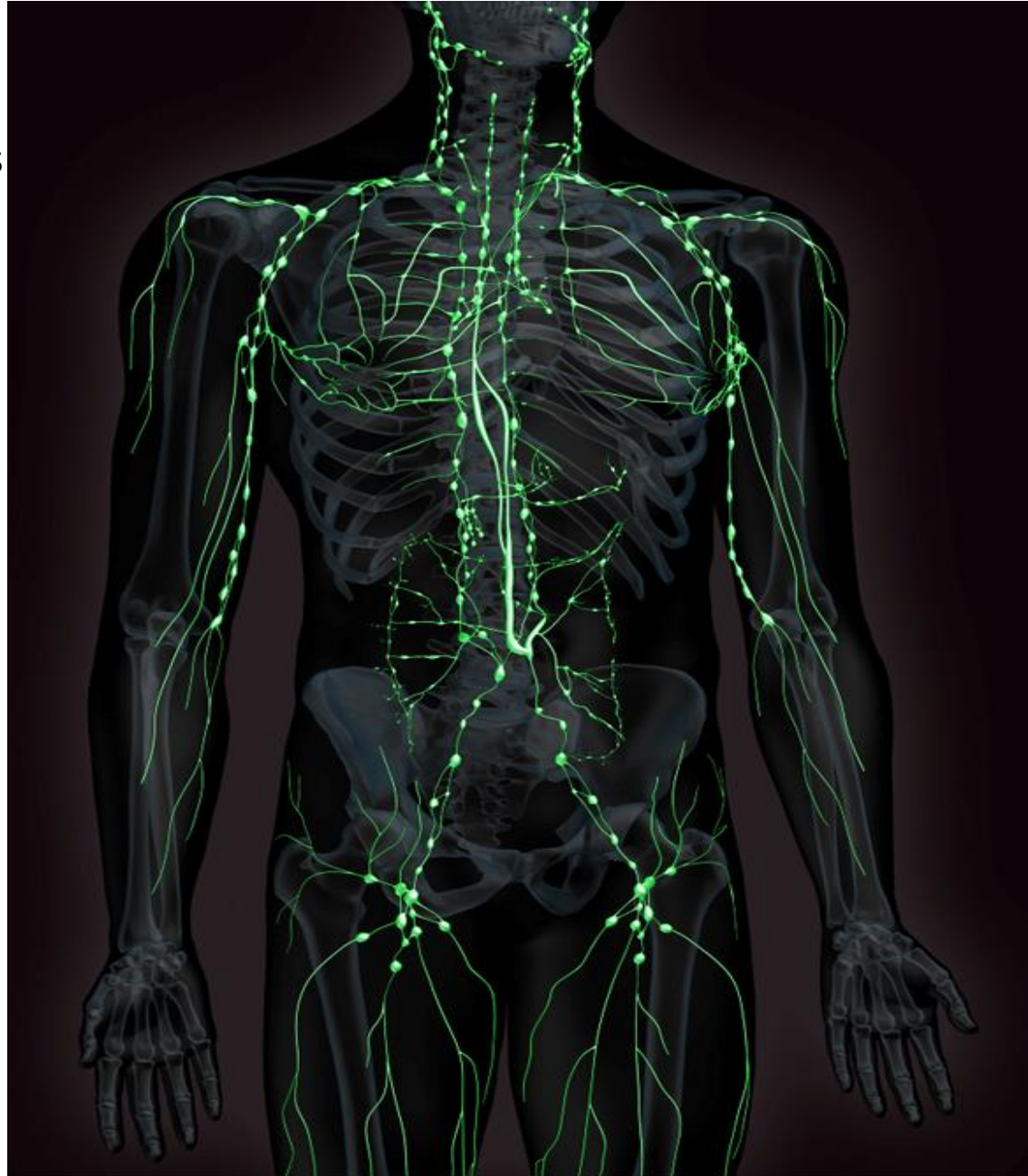
Each cell is packed tightly next to another of the same type to make a specific tissue type (e.g. muscle, bone, nerves)

Cancer cells occur where the cells replicate too quickly and don't die in the normal way



Our lymphatic system

- Similar to blood circulation, but uses much narrower tubes
- Carries away harmful bacteria, dead / abnormal cells etc
- Collects in regional lymph nodes where they are filtered out
- Cancer cells collect in these nodes and continue to grow
- The cancer cells are then transported to other parts of the body



Risk factors

- Smoking – damages cell DNA
- Inherited – faulty genes pass down the family line
- Lifestyle – obesity, poor diet, alcohol
- Viruses – trigger genetic changes
- Age – reduced immune system and long term exposure to carcinogens

Symptoms of cancer

- Unusual lump, or change to an existing one
- Breathlessness
- Difficulty swallowing
- Night sweats
- Weight loss
- Blood in faeces or urine
- Unexplained long-term pain

Benign vs Malignant

- Benign tumours tend to be:
 - Mildly abnormal cells
 - Grow slowly
 - Do not spread round the body
- Malignant tumours tend to be:
 - Markedly abnormal cells
 - Grow quickly
 - Spread into and destroy surrounding cells/tissue
 - Spread around the body

Cancer staging and grading

- TNM staging classification

T = tumour size (T0, Tis then T1 -> T4)

n.b. T1 bladder cancers are not usually invasive

N = nodes involved (N0 -> N3)

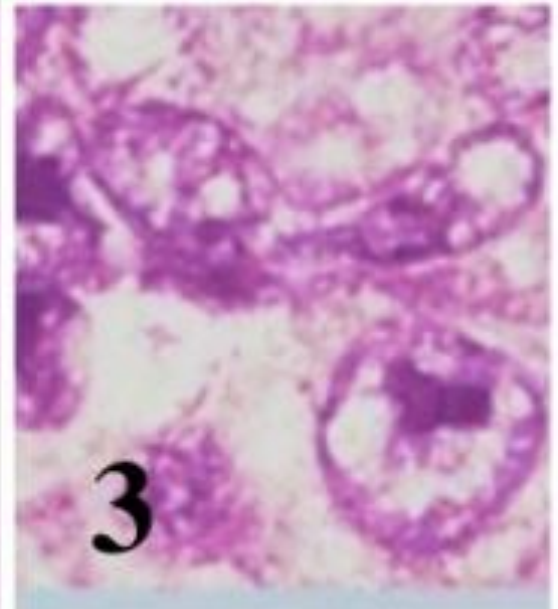
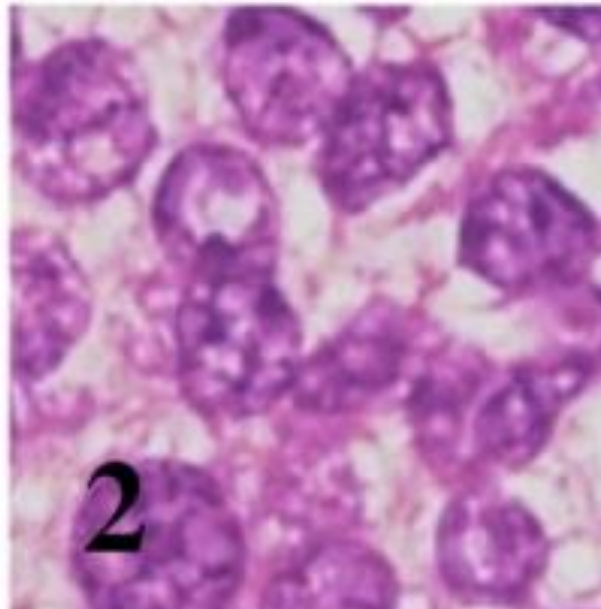
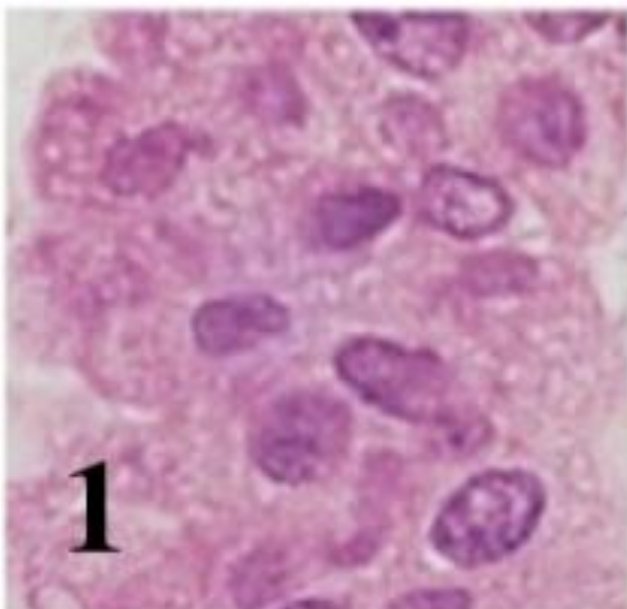
M = metastases (M0 or M1)

(Some companies pay lower amounts for in-situ cancers)

e.g. T1N0M0, T3NxM1, pT2N1M0, TisN0M0

- Grading classification

- Grade 1 – well differentiated
- Grade 2 – moderately differentiated
- Grade 3 – poorly differentiated



ABI cancer definitions

- The first ABI definition of cancer in 1999:

“A malignant tumour characterised by the uncontrolled growth and spread of malignant cells and invasion of tissue. The term cancer includes leukaemia and Hodgkin’s disease but the following are excluded:

- All tumours which are histologically described as pre-malignant, as non-invasive or as cancer in situ.
- All forms of lymphoma in the presence of any Human Immunodeficiency Virus.
- Kaposi's sarcoma in the presence of any Human Immunodeficiency Virus.
- Any skin cancer other than malignant melanoma.”

ABI cancer definitions

- Most recent ABI definition of cancer 2014:

“Any malignant tumour positively diagnosed with histological confirmation and characterised by the uncontrolled growth of malignant cells and invasion of tissue. The term malignant tumour includes leukaemia, sarcoma and lymphoma except cutaneous lymphoma (lymphoma confined to the skin). For the above definition, the following are not covered:

- All cancers which are histologically classified as any of the following:
 - pre-malignant;
 - non-invasive;
 - cancer in situ;
 - having borderline malignancy; or
 - having low malignant potential;
- All tumours of the prostate unless histologically classified as having a Gleason score of 7 or above or having progressed to at least TNM classification T2bN0M0.
- Chronic lymphocytic leukaemia unless histologically classified as having progressed to at least Binet Stage A.
- Any skin cancer (including cutaneous lymphoma) other than malignant melanoma that has been histologically classified as having caused invasion beyond the epidermis (outer layer of skin). “

ABI cancer definitions

- Advances in medical detection and treatment
 - Tumours found much earlier so better response to treatment
 - Increased numbers of claims
 - More effective treatment
 - Moving away from 'critical' illness

Prostate cancer example

2002 definition exclusion “All tumours of the prostate unless...progressed to at least TNM classification T2N0M0”

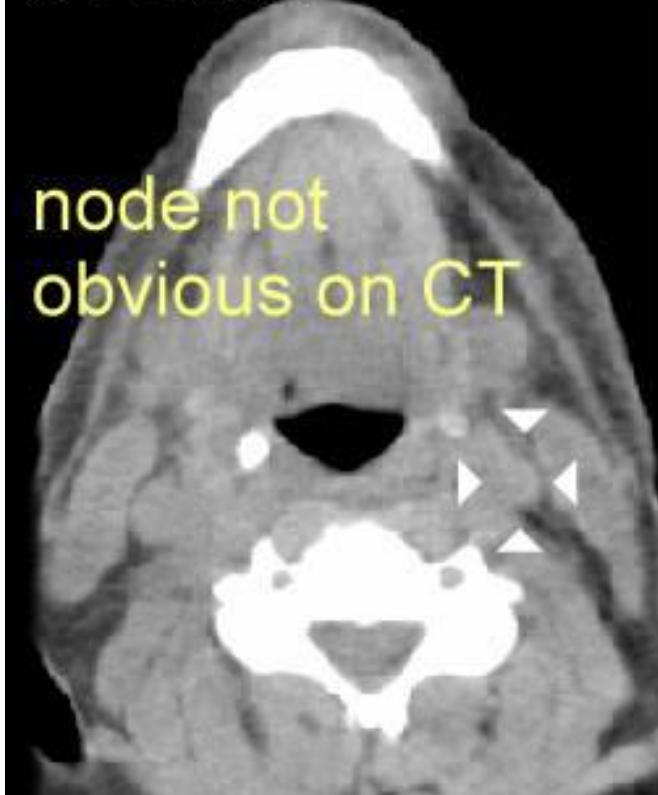
2006 definition exclusion “All tumours of the prostate unless...progressed to at least clinical TNM classification T2N0M0”

Investigations

- Scanning
 - USS (ultrasound scan)
 - MRI (magnetic resonance imaging)
 - CT (computerised tomography)
 - PET (positron emission tomography)

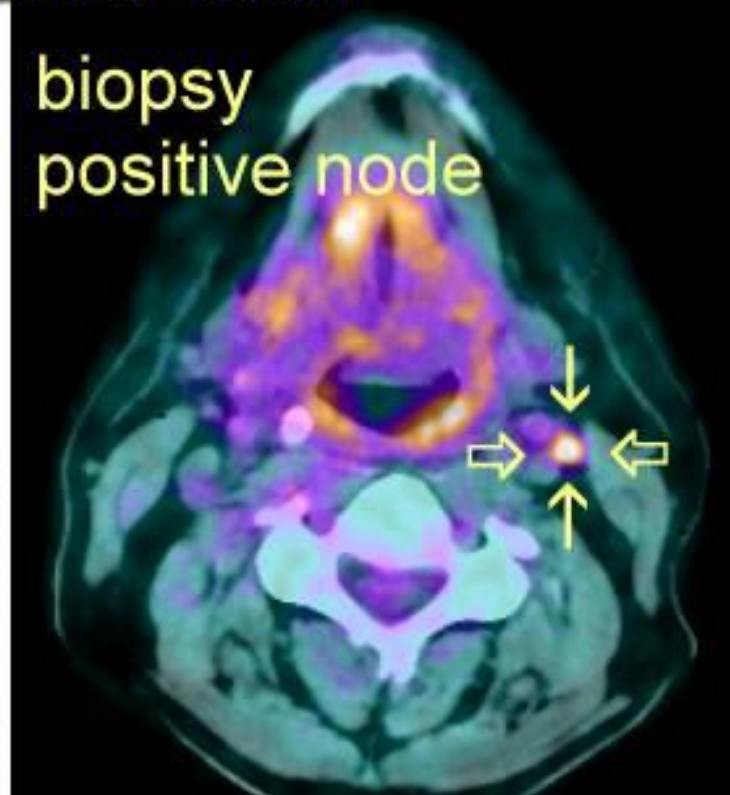
CT Scan

node not
obvious on CT



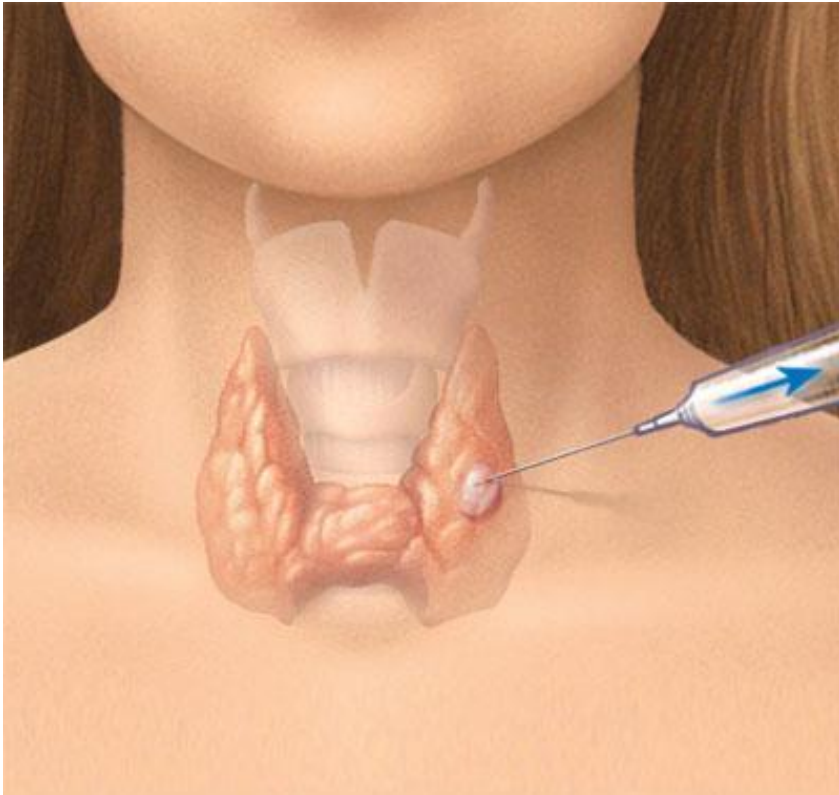
PET Scan

biopsy
positive node

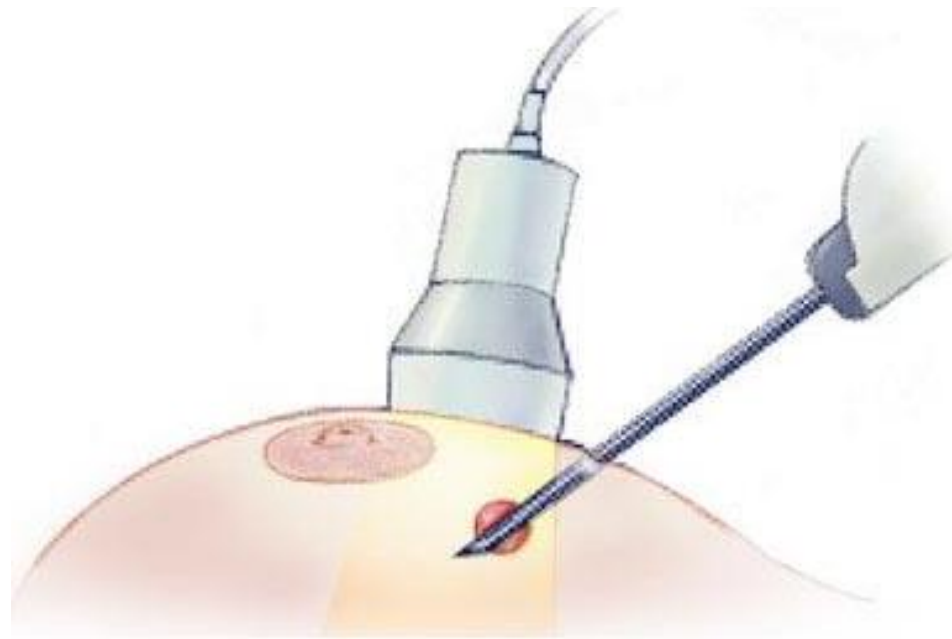


Investigations

- Fine needle aspiration (FNA)



- Core needle biopsy



Investigations

- A biopsy is the removal and microscope examination of a suspected cancer.
- The tissue is sliced, dyed, and sometimes frozen.
- The **histology report** will describe
 - The appearance of the cells within each 'slice'
 - Whether there is lymphovascular invasion
 - Whether the margin, or edge, of the biopsy, is clear of cancer cells.
 - The grade of the cancer
 - The mitotic rate of the cancer cells
- If the margins were not clear, a wider excision biopsy will be performed.



Assessing CI claims

- Histology reports
 - OBTAIN the histology report for the patient if available
 - ENSURE the report you get is at least the main (core) biopsy
 - DO NOT be confused with an earlier FNA that might be negative
 - DO NOT be confused with later wider excision biopsies
 - DO NOT be confused with reports which contain elements of a cancer that doesn't qualify for CI e.g.

Microscopic Appearance:

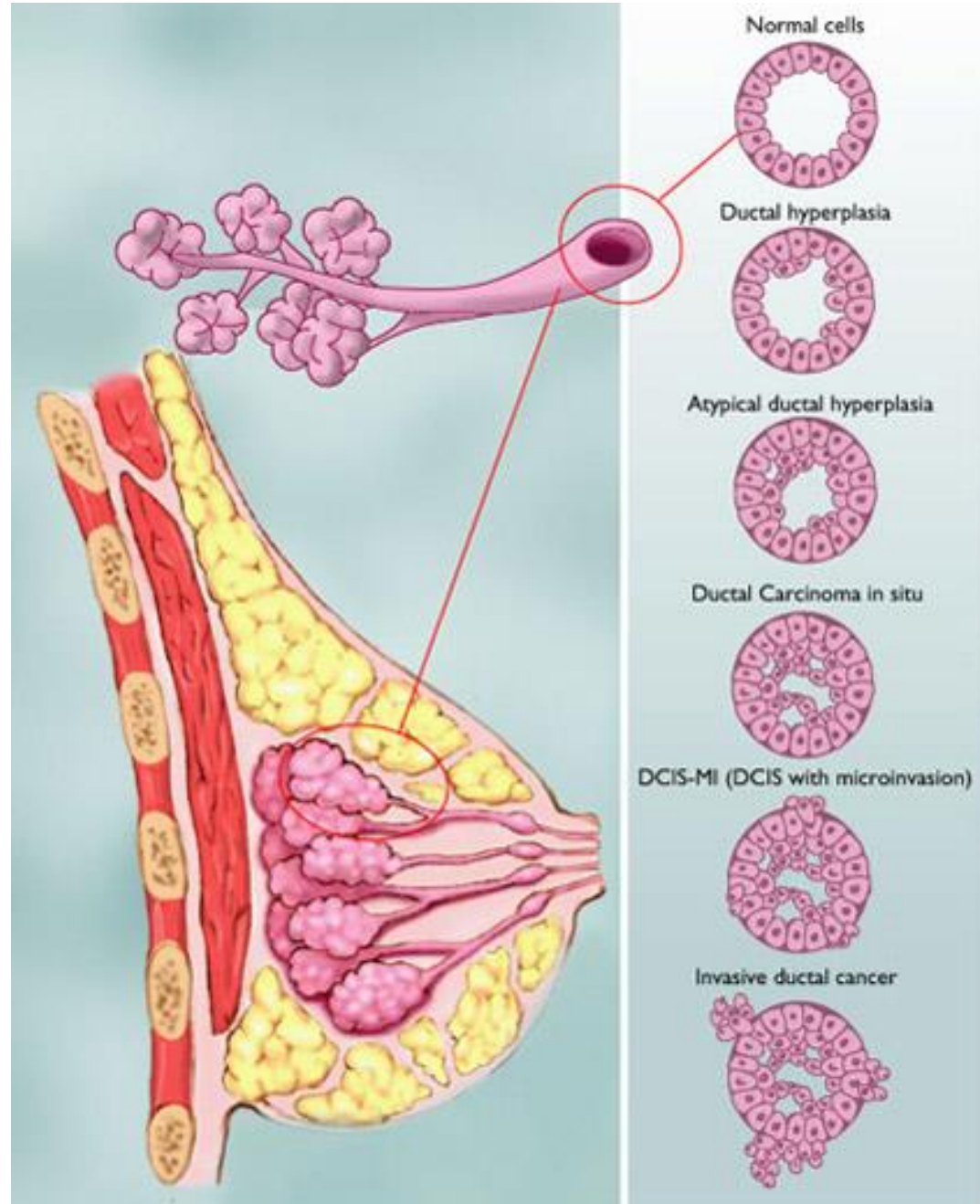
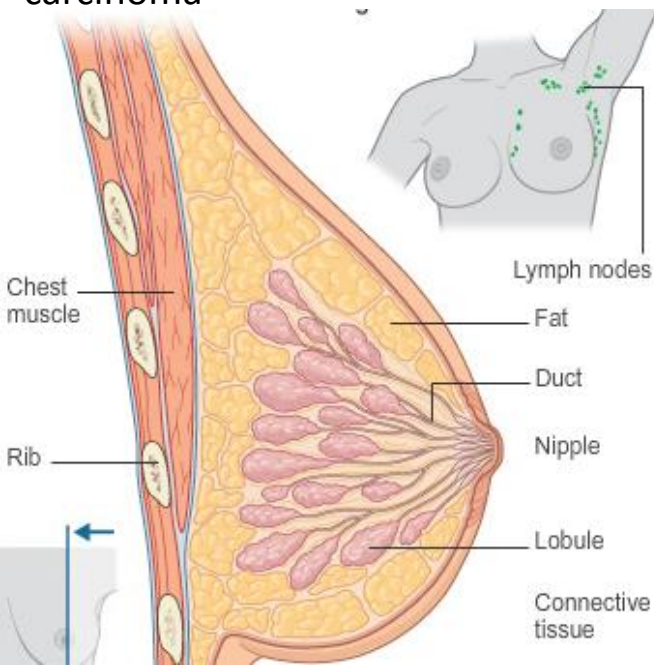
The breast core biopsy reveals infiltrating ductal carcinoma (predicted grade 2) and focal ductal carcinoma in situ (DCIS), of probable high nuclear grade. There is no lymphovascular invasion seen. Focal calcification is also noted.

Breast cancer

- Mammograms are routinely done every 3 years from aged 50 in the UK
- Ultrasounds are performed on younger patients with suspicious lumps

Most cancers occur in the milk ducts = ductal carcinoma

Or in the tiny lobules that make milk = lobular carcinoma



Breast cancer staging

TNM classification system

- T1 <2cm (T1a <5mm; T1b 5-10mm; T1c 1-2cm)
- T2 2-5cm
- T3 >5cm
- T4 extended into chest wall

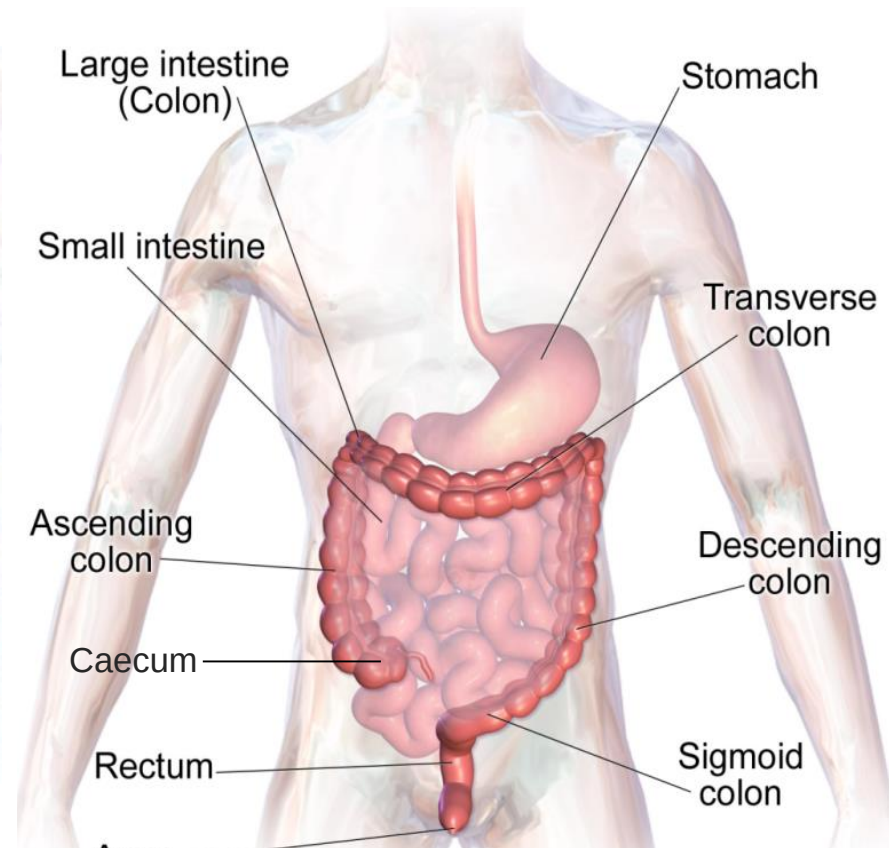
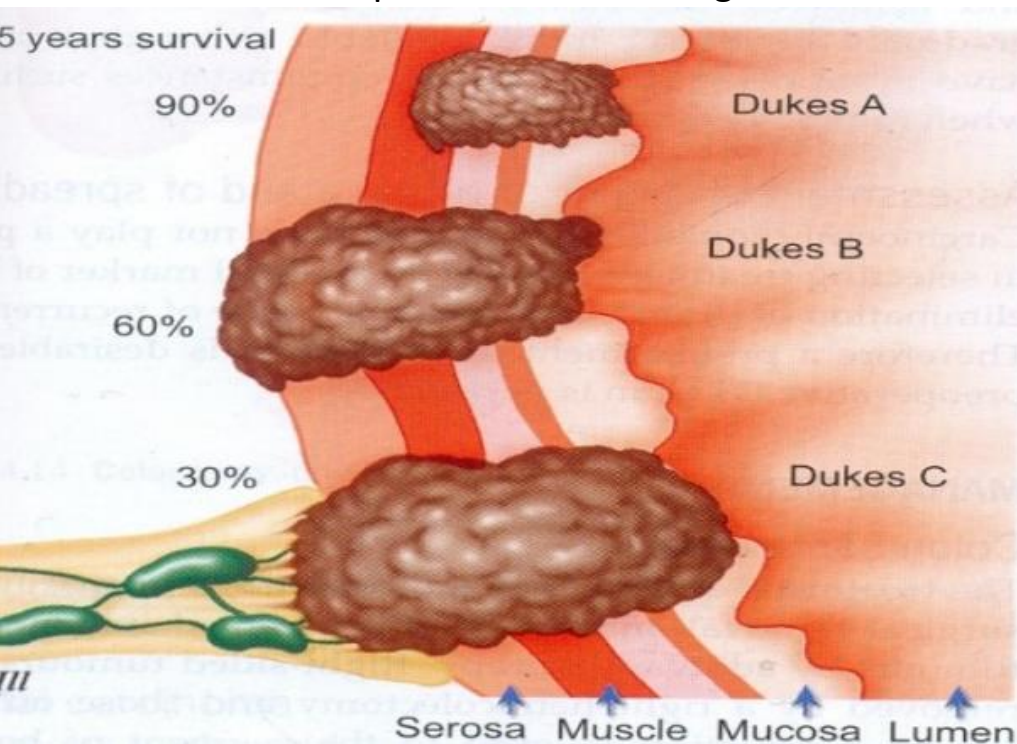
Group Staging system

- Determined by
 - Tumour size
 - Level of invasiveness
 - Lymph node involvement

Stage	Description
0	In situ only
I	T1N0M0 T1NmiM0
II	TxN1M0 T1N1M0 T2N1M0 T3N0M0
III	TanyN2M0 T3NmiM0 T3N1M0
IV	TanyNanyM1

Bowel cancer

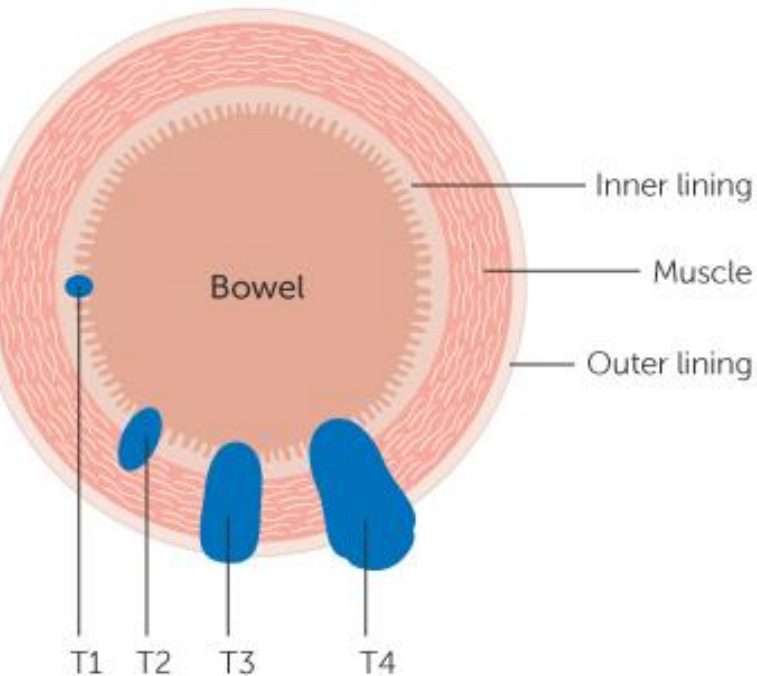
- Sigmoidoscopy or colonoscopy used to diagnose bowel cancer
- Most tumours are adenocarcinomas (arise from epithelial tissue)
- Classifications are TNM, Group Staging, and Dukes.
 - Dukes A = confined to innermost lining of colon/rectum or slightly into muscle
 - Dukes B = grown through the muscle layer of the colon/rectum
 - Dukes C = spread to at least 1 lymph node
 - Dukes D = spread to another organ



Bowel cancer - TNM and Group staging systems

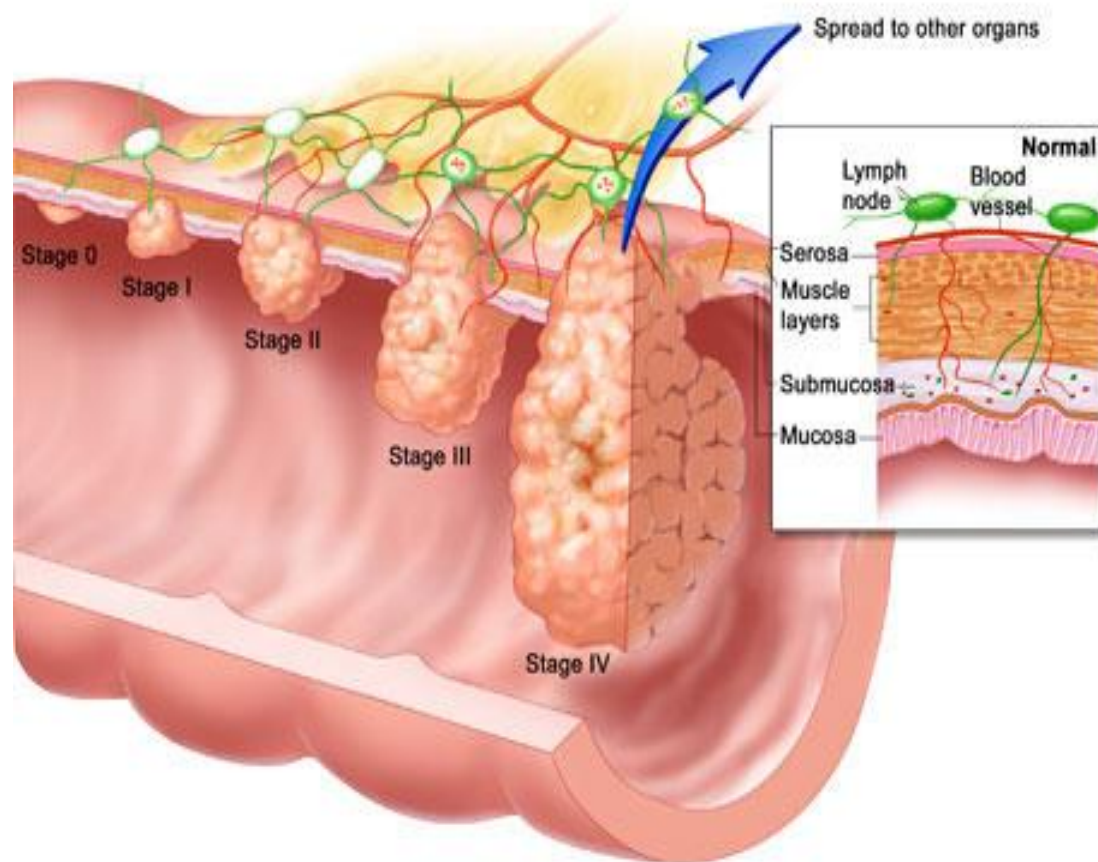
T classification:

- T1 = invasion not as far as mucosa
- T2 = penetrated muscle layer
- T3 = reached serosa
- T4 = beyond bowel wall



Group Staging:

- Stage 0 = in-situ
- Stage 1 = growth into lining or muscle (i.e. T1 or T2)
- Stage 2 = no lymph node involvement (i.e. T3N0 or T4N0)
- Stage 3 = lymph node involvement
- Stage 4 = metastases



Skin cancer

➤ Basal cell carcinoma & Squamous cell carcinoma

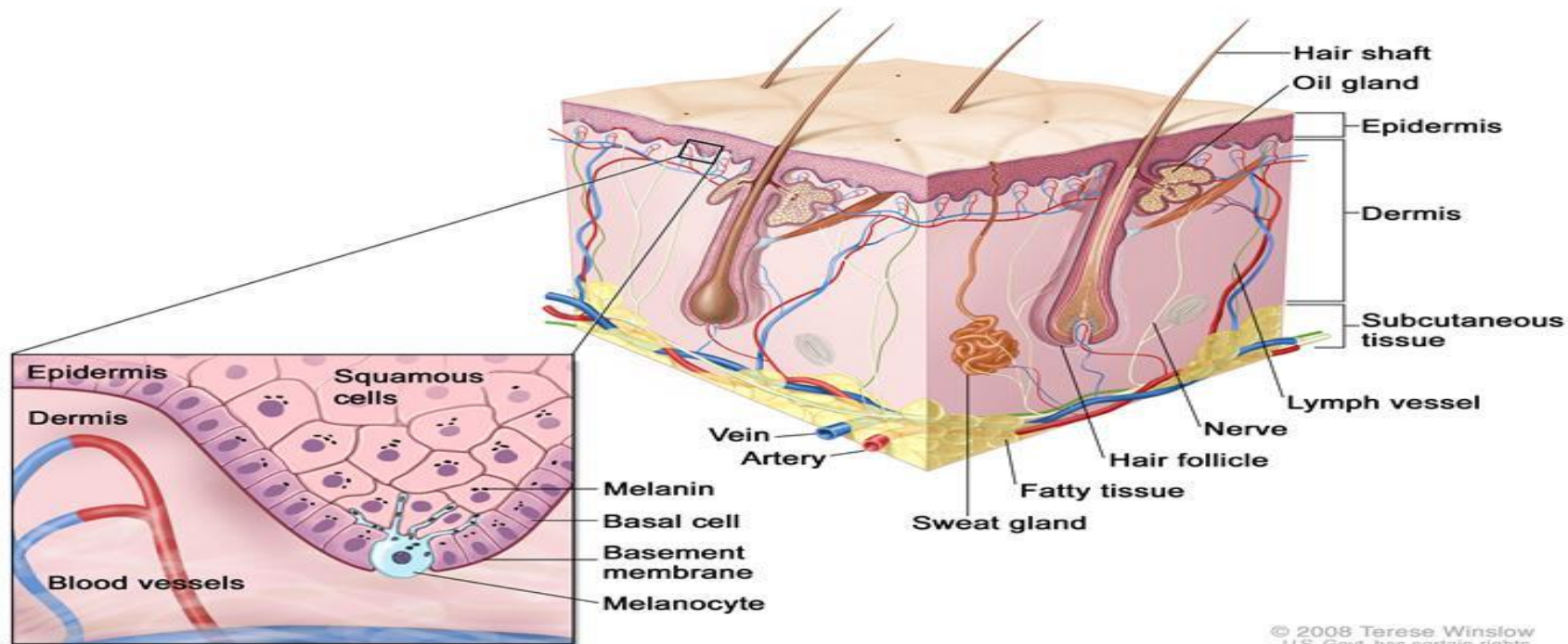
- These are both excluded from policy definitions
- They don't spread and are slow growing

➤ Melanoma

- Arise from melanin producing cells
- Grow quickly
- Spread round the body if not treated

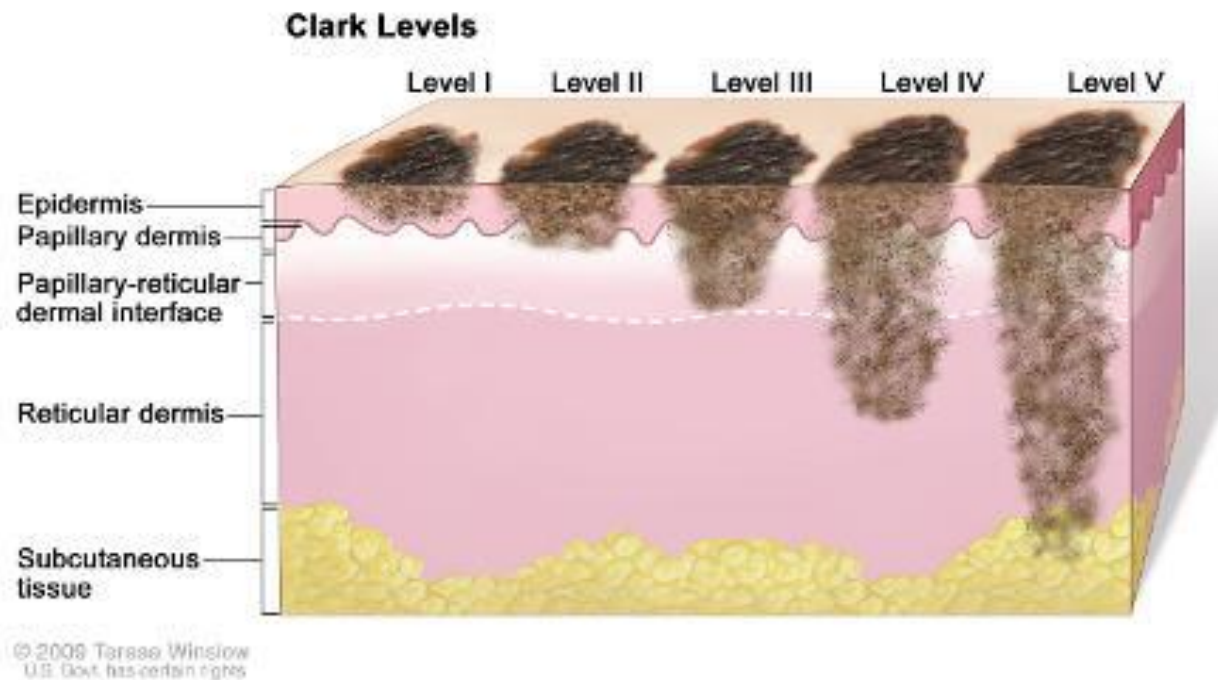
○ Histology

- Superficial spreading malignant melanoma
- Ulceration – describes the surface layer of the tumour
- Breslow thickness



Skin cancer

- TNM classification
 - T1 invasion <1mm
 - T2 1-2mm
 - T3 2-4mm
 - T4 >4mm
- Clark level:



ABI definition exclusion: Any skin cancer other than malignant melanoma that has been histologically classified as having caused invasion beyond the epidermis.

Skin cancer – all staging types

Group Stage	TNM	Breslow	Description
0	Tis N0 M0		In situ melanoma only Clark level I
IA	T1a N0 M0	<1mm	No ulceration involved, mitosis <1/mm ² Clark II/III*
IB	T1b N0 M0	<1mm	Ulceration involved, mitosis ≥1/mm ² Clark IV/V*
	T2a N0 M0	1-2mm	No ulceration involved
IIA	T2b N0 M0	1-2mm	Ulceration involved,
	T3a N0 M0	2-4mm	No ulceration involved
IIB	T3b N0 M0	2-4mm	Ulceration involved.
	T4a N0 M0	>4mm	No Ulceration involved
IIC	T4b N0 M0	>4mm	Large melanoma with ulceration
III	T-N-M0		Lymph nodes are involved, irrespective of how many.
IV	T-N-M1		Distant metastases.

Case study 1

- Female 45
- Lump discovered on routine mammogram
- Dx in-situ ductal carcinoma on FNA
- Went on to core biopsy & mastectomy – final histology showed T1N0M0 ductal carcinoma with areas of ductal carcinoma in-situ
- Claim decision?

Case study 2

- Male 32 – skin lesion

In these superficial, curetted fragments, the epidermis is thickened and shows widespread, Pagetoid intra-epidermal spread of atypical melanocytes, accompanied by occasional, atypical, melanocyte nests at the dermo-epidermal junction, in keeping with melanoma in situ.

In one level, there is a single, small group (around 20 cells) of atypical melanocytes within the papillary dermis, interpreted as early vertical growth phase melanoma, Breslow thickness 0.3mm, Clark level 2.

No ulceration, regression, lymphovascular/perineural invasion, mitoses or microsatellites are seen in these curettings. Infiltrating lymphocytes are absent. Completeness of excision cannot be assessed in these fragments.

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Case study 2

- Male 32

Microscopy

SPECIMEN A

The sections show fragments of skin containing part of a compound predominantly junctional melanocytic lesion. In the epidermis there is lentiginous proliferation and focal nesting of atypical melanocytes with pagetoid spread. Ulceration is not seen. Tumour cells extend into neighbouring adnexal structures. In the dermis available for evaluation there is evidence of a tiny pre-existing naevus. In one of the sections there is a very small nest of atypical melanocytes that appears to be in the papillary dermis. This however, is more likely to represent a nest of melanocytes in an elongated rete that has been sectioned transversally rather than early invasion (0.3 mm). No regression is seen.

Conclusion: Melanoma in-situ and evidence of a pre-existing naevus. Re-excision is indicated.

Case study 3

- Male 58
- Histology has been misplaced but Consultant report details Stage 2 adenocarcinoma of the bowel. No family history and sudden onset of symptoms.
- Decision?

Questions, or lunch...?

